MIL-P-157A (MR)
2 APRIL 1963
SUPERSEDING
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MILITARY SPECIFICATION

PAPER, ONIONSKIN

1. SCOPE

1.1 This specification covers one type of onionskin paper for use in ammunition items (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

UU-P-31 — Paper; General Specifications and Methods of Testing.

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 — Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Freight Classification Committee, 202 Union Station, Chicago 6, Illinois.)

3. REQUIREMENTS

3.1 **Material.** The product shall consist of rag stock and bleached or partially bleached chemical wood pulp, without the admixture of ground or unbleached fibers, constructed in such a manner as to meet the requirements of this specification.

3.2 **Form.** Unless otherwise stated in the contract or order, the onionskin paper shall be in the form of sheets (see 3.5).

3.3 **Color.** The color of the paper shall be white to buff. Conformance shall be determined in accordance with 4.5.1.

3.4 **Finish.** The finish of the paper shall be glazed. Conformance shall be determined in accordance with 4.5.2.

3.5 **Sheet size.** The sheets of onionskin paper shall be 22 $\pm$ $\frac{1}{16}$ by 17 $\pm$ $\frac{1}{16}$ inches in size, unless otherwise specified in the contract or order (see 6.2).

FSC 8135

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3.6 Property values. The onionskin paper shall conform to the requirements listed in table I, when tested in accordance with the applicable test procedures of paragraph 4.5.

TABLE I

Property	Requirements	Test paragraph
Thickness, inch	0.0015 ± 0.0005	4.5.4
Weight, pounds per 1000 sheets	16 ± 1.0	4.5.5
Bursting strength, points, minimum	12	4.5.6
Rag content, percent minimum	30	4.5.7
Ash, percent, maximum	2.0	4.5.8
Acidity (as SO_3) or alkalinity (as NaOH), % maximum	0.20	4.5.9

3.7 **Workmanship.** The onionskin paper shall be uniform in quality and construction, free from foreign matter, cuts, holes, wrinkles, chafed spots, or other defects which might impair the usefulness of the material for the purpose intended. The sheets shall be evenly stacked and shall not stick together to the extent that the material will be damaged when the sheets are separated.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 **Lot.** A lot shall consist of all the onionskin paper which has been manufactured from the same types and proportions of ingredients, by the same process, by one manufacturer and submitted for acceptance at one time.

4.3 **Sampling.** For purposes of sampling, the unit shall be one stack or package of 1000 sheets. Not more than 20 sheets shall be examined from any one stack or package of sheets except when required for determining the count of sheets.

4.3.1 *Sampling for end item examination.* Samples for end item examination shall be selected in accordance with inspection level I–5 of MIL–STD 105 and at the acceptable quality levels shown in the classification of defects (see 4.4.1).

4.3.2 *Sampling for inspection of preparation for delivery.* Samples for inspection of packaging, packing and marking shall be selected in accordance with inspection level L–7 of MIL–STD–105 and at the acceptable quality levels shown in 4.4.2.

4.3.3 *Sampling for tests.* Sample units for testing shall be selected in accordance with inspection level L–4 of MIL–STD–105, acceptance number zero. The specimen sample shall consist of 25 sheets (each 22 by 17 inches) to be taken from each sample unit selected for test purposes.

4.4 Inspection.

4.4.1 *End item examination.* The samples selected as in 4.3.1 shall be examined in accordance with the classification of defects and at the acceptable quality levels shown in tables II and III.

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TABLE II. *Classification of defects*
(Acceptable Quality Level 2.5 percent defective)

Examine	Defect
Form	Not in sheet form or as specified.
Cleanliness	Not clean. Material shows dirt, sand, grit, oil or other foreign matter.
Blocking	Adjacent sheets stick together to the extent that they are damaged when separated.
Workmanship	Any cut, hole, wrinkle, chafed spot of other defect affecting serviceability.

TABLE III. *Classification of defects*
(Acceptable Quality Level 4.0 percent defective)

Examine	Defect
Stacking	Not evenly stacked.
Sheet count	Average sheet count of stacks varies from 1000 by more than 25 sheets.

4.4.2 *Inspection of preparation for delivery.* The packaging, packing and marking of the onionskin paper shall be examined to determine conformance to the requirements of section 5 of this specification, and as required by specifications referenced therein. The acceptable quality level shall be 4.0 percent defective.

4.5 Test procedures.

4.5.1 *Color.* Compliance with color requirements shall be determined by visual examination.

4.5.2 *Finish.* Compliance with finish requirements shall be determined by visual examination.

4.5.3 *Sheet size.* Three sheets shall be measured by any suitable means giving precision to $\frac{1}{16}$ inch for length and width determinations. The average shall be reported as the length and width values.

4.5.4 *Thickness.* Compliance with thickness requirements shall be determined in accordance with method 173 of Federal Specification UU–P–31.

4.5.5 *Weight.* Compliance with weight requirements shall be determined in accordance with method 110 of Federal Specification UU–P–31.

4.5.6 *Bursting strength.* Compliance with bursting strength requirements shall be determined in accordance with method 112 of Federal Specification UU–P–31.

4.5.7 *Rag content.* Rag content shall be determined in accordance with method 300 of Federal Specification UU–P–31.

4.5.8 *Ash content.* Ash content shall be determined in accordance with method 202 of Federal Specification UU–P–31.

4.5.9 *Acidity (as SO_3) or alkalinity (as NaOH).* Acidity or alkalinity shall be determined as follows: (2 determinations shall be made). Sufficient square shaped pieces (not over $\frac{1}{4}$ inch in length or width) shall be cut from material as received so as to provide approximately 5 grams of air dried shredded material. This material shall be placed in a 500 milliliter Pyrex Erlenmeyer (or equivalent) flask and 250 ml. of distilled water added. To negate the tendency of the pieces to float on the surface, the water is added gradually and the flask well shaken. After the water is added, affix to the flask a foil covered, rubber stopper through which a glass tube about 30 inches in length has been inserted to serve as a condenser. Alternately a 500 ml. Pyrex (or equivalent)

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Erlenmeyer flask with a ground glass top and a water cooled condenser with a ground joint to fit the flask may be used. Place the flask and contents on a hot plate and reflux gently for 1 hour with occasional shaking to ensure that all pieces are immersed in the water. Immediately pour the contents of the flask on a Buchner funnel (without other filtering medium) receiving the solution in a 400 ml. Pyrex (or equivalent) beaker. Rinse twice using 25 ml. of hot distilled water each time. Cover the beaker with a watch glass and cool to room temperature rapidly. Run a blank in the same manner described for the actual determination using 250 ml. of distilled water without the sample. Cool the water to room temperature and with a calibrated pH meter determine the pH value of the blank solution. Titrate the sample extract to the pH value of the blank using 0.01N HCL solution, or 0.01N NaOH solution whichever is appropriate. Calculate any acidity in terms of SO_3 or alkalinity in terms of NaOH by the following formulae. Report individual results of all testing.

$$\text{Percent } SO_3 = \frac{T \times N \times 4}{W}$$

where:

T = ml. NaOH required to titrate the extract to the pH of the blank

N = Normality of NaOH solution

W = Weight (in grams) of test specimen

or

$$\text{Percent NaOH} = \frac{T \times N \times 4}{W}$$

where:

T = ml. of HCL required to titrate the extract to the pH of the blank

N = Normality of HCL solution

W = Weight (in grams) of test specimen

5. PREPARATION FOR DELIVERY

5.1 **Packaging.** Packaging shall be level C or as specified in the contract or order.

5.1.1 *Level C.* Stacks of onionskin paper (see 4.3) shall be packaged in a manner which will afford adequate protection against deterioration and damage during shipment from the supply source to the first receiving activity for immediate use or for controlled humidity storage.

5.2 **Packing.** Packing shall be level C or as specified in the contract or order.

5.2.1 *Level C.* Packing shall be accomplished in a manner which will afford adequate protection against damage during shipment within the Continental United States to the first receiving activity. This level as a minimum will conform to applicable carrier rules and regulations.

5.3 **Marking.** Unless otherwise specified by the procuring activity, individually packaged stacks of sheets and exterior containers shall be marked for shipment in accordance with Standard MIL-STD-129, and as follows:

"Paper, Onionskin"

"Specification MIL-P-157A(MR)"

Size

Quantity

Contract or order number

Weight

Name and address of manufacturer

Contract or order number

Month and year of manufacture

6. NOTES

6.1 **Intended uses.** The onionskin paper covered by this document is primarily intended for use in ammunition items for such applications as separator discs to prevent explosives dust from one component from

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contacting another explosive or other such component.

6.2 **Ordering data.** Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Sheet size, if other than 22 by 17 inches (see 3.4).

Notice. When Government drawings, specification, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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